

Problems Encountered & Solutions

Group 22 - SafeX Educational Learning Platform - Week 3 - Muhammad Abdullah

Problem	Cause	Solution
SSMS refused to connect to local SQL Server.	SQL certificate chain not trusted; SSMS 20 defaults to Encrypt=Mandatory.	In Connect to Options to Connection Properties, set Encrypt=Optional and enabled Trust Server Certificate.
React could not call ASP.NET API.	Different origins (5173 vs 5000) triggered a CORS block.	Added AddCors policy 'frontend' in Program.cs allowing http://localhost:5173.
YouTube search returned no duration or view count.	search.list only returns snippet data.	Made a second videos.list call with contentDetails,statistics for the returned IDs.
Duration came back as PT4M13S string.	YouTube uses ISO-8601 duration format.	Wrote parseDuration() to convert PTxHxMxS into seconds for display.
Quota almost drained during testing.	Every keystroke fired a search.	Added 500 ms debounce on the search input and disabled auto-fire.
Re-importing the same video threw a unique key error.	YouTubeld column is UNIQUE.	Switched the insert to an upsert (update if exists, insert if new).
API key almost got committed to Git.	appsettings.json was tracked.	Added appsettings.Development.json and .env.local to .gitignore; documented setup steps in README.

Lessons learned

- Always check the API cost per call before wiring it to a live input.
- Keep secrets out of source control from day one; retrofit is painful.
- CORS, HTTPS, and cert trust are the top 3 first-time full-stack blockers.
- Upserts are safer than inserts when integrating third-party IDs.